

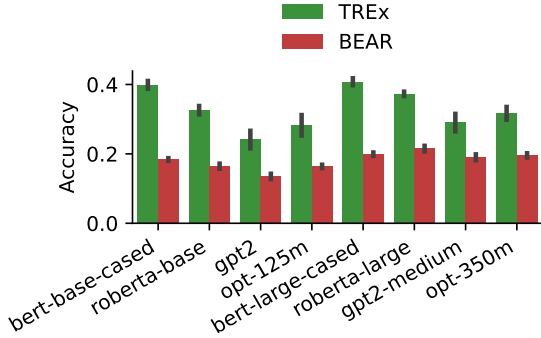


Figure 5: Comparative analysis of model performance on identical subsets of relations and templates in the T-REx (LAMA) and BEAR datasets using the log-likelihood based evaluation.

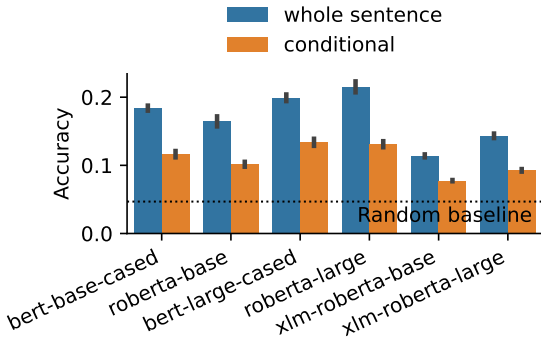


Figure 6: Aggregated accuracy (measured on BEAR) when using the sum over all tokens in the complete statement vs. answer-tokens only.

edge encoded within its parameters.

We also find that the performance disparity evaluated on BEAR’s relations is less pronounced than across the same subset of T-REx’s relations. Such difference may indicate that models pre-trained on Wikipedia (like BERT) have an advantage on LAMA over those not trained on Wikipedia (such as GPT2) due to a potential train/test data overlap. For a detailed performance comparison on a per-relation basis, refer to Figure 8 in the Appendix. For example, bert-base-cased achieves a very high performance on T-REx’s MANUFACTURER relation (ID: P176) but a significantly lower score on the corresponding BEAR subset.

5.2.1 Ablations

Conditional scores. To compute the pseudo log-likelihood of a statement in masked language models, one forward pass per token is required. However, masking only the tokens that are part of the an-

swer would significantly reduce the required computation. One may expect that the representation of the answer tokens may be sufficient to predict the likelihood of each answer. Still, our experiments (see Figure 6) indicate there is a significant⁶ drop in performance when using the conditional score (i.e., the score of the answer tokens conditioned on the context instead of the pseudo log-likelihood of the complete sentence). Coincidentally, for entities represented by a single token, the conditional score matches the score predicted for the [MASK] token, similar to the approach first used in the LAMA probe.

Sum vs. mean of the log-likelihood. We investigate how performance varies by scoring the sentences using both sum and mean reduction methods. We observe that scoring the sentence by computing the mean over the token log-likelihoods tends to yield inferior performance for the probe. Figure 7 (in Appendix A) illustrates the results of this ablation study.

To understand why this might be the case, consider how the word ‘souvenir’ can be broken down into subword tokens: ‘so’, ‘##uven’, and ‘##ir’.⁷ The first token ‘so’ may have a relatively small log-likelihood, ‘##uven’ a bit higher (since it’s conditioned on the first token), and finally ‘##ir’ a log-likelihood of almost 0 since there are few other ways to continue the statement. In contrast, the single-token word ‘gift’ may have a medium log-likelihood, which may still be lower than the average of the three-token word ‘souvenir’. This example illustrates that the mean can inflate the predicted probabilities of longer answers or sentences.

We suggest summing the tokens’ log-likelihoods for both masked and casual language models in future experiments.

6 Conclusion

We presented BEAR, a relational knowledge probe applicable to both causal and masked LMs. Since our proposed approach imposes no restrictions on the evaluation data, we created a large evaluation dataset that addresses issues of answer skews, domain and template bias, and the correctness of facts identified by ourselves and prior work. We publicly release BEAR for use by the research community.

⁶P-value of 1.1×10^{-11} ; using a Student’s t-test for paired samples

⁷The example was introduced by Kauf and Ivanova (2023) to make a different point but is also relevant here.

Limitations and Risks

The knowledge probe we present in this paper follows the approach of earlier probes and, as such, tests only for factual, relational knowledge. This kind of knowledge includes classic relationship types such as the place of birth of persons, their time of birth, the genre of works of art, etc. However, one might be interested in testing a model for other types of more general commonsense knowledge, such as physical reasoning or general properties of concepts. Our probe does not test for such kinds of knowledge.

Furthermore, even though we devised heuristics to ensure that entities in BEAR are common enough to appear on Wikipedia pages of many different languages and record a certain number of page views, there remains a likely bias towards entities overrepresented on Wikipedia, giving an advantage to LMs trained on Wikipedia rather than more general corpora.

We see few risks in the BEAR probe itself but caution that knowledge probing is often used to assist in the research and development of LMs. As such, BEAR may contribute to developing LMs that malevolent actors might misuse.

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